Chapter 7: Asset Prices, Uncertainty and a General Ricardian Irrelevance Result

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1 Overview

The last chapter did not present explicit notation for uncertainty, i.e. keeping track of different states of the world following the tradition in the New Keynesian literature (see e.g. Woodford (2003) and Galí (2019)). While we will follow this tradition in other chapters, we will here develop more explicit notation for uncertainty. Overall, this chapter is a bit dense on notation, and thus may be a bit taxing on the reader's patience. The objective is to give a brief and condensed overview of the foundations of various objects that we will use heavily later on, which otherwise may seem mysterious, such as the stochastic discount factor, as well as to provide a condensed proof for the transversality condition, following the exposition of Woodford (2003). The model of this chapter is the same as in the last chapter, a representative household living in an endowment economy, except that we allow for more detailed description of uncertainty and a richer menu of assets. The concepts introduced, however, apply to much richer models.

The purpose, among other things, is to convince you of the derivation of the consumption Euler equation, where we appended the conditional expectation operator. Another objective is to introduce you to the stochastic discount factor, as well as the pricing of various assets. This investment allows us to generalize the Ricardian equivalence proposition in last chapter. There, we showed that the determination of short-term real interest rates was independent of the time path of taxes and debt. That model, however, only allowed for one-period, risk-free debt. Here, instead, we allow the government to purchase or sell any type of security. This could include, for example, long-term government debt or even common stocks. This could also include securities that are risky. Furthermore, we allow the government to hold arbitrary positive or negative quantities of these securities. The bottom line is that these purchases will have no effect on the prices of the assets purchased or sold. This insight will be fundamental to

understanding the irrelevance result of Eggertsson and Woodfoord (2003), who showed that Quantitative Easing has no effect in the New Keynesian model we will develop later on.

2 Stochastic structure

The notation and assumptions about the stochastic structure differ across papers and textbooks and are a matter of convention. I choose a notation convention I suspect is the most common and which is adapted in many textbooks and articles, see for example Ljungqvist and Sargent (2020).

The variable s_t denotes a realization of a stochastic event $s_t \in S$ which corresponds to, for notational simplicity, a finite number of possibilities at time t. The set of possible events, S, could for example be [good weather, bad weather]. Let the history of events up to time t be denoted by a superscript t, i.e., $s^t = [s_t, s_{t-1}, ..., s_0]$. Thus history for the last two days, for example, could be $s^t = [s_t = \text{good weather}, s_{t-1} = \text{bad weather}]$. Let the probability of observing a particular sequence of events s^t be denoted $\pi(s^t)$.

Think of date 0 as the initial period in which the state of the world is observed, accordingly s_0 is non-stochastic and $\pi(s_0) = 1$ for a particular $s_0 \in S$ (but there may be many possible initial conditions). Let us suppose that history s^{t-1} is followed by history s^t Let us denote the conditional probability of observing history s^t given history s^{t-1} by $\pi(s^t/s^{t-1})$. It then follows from Bayes's law that[1]

$$\pi(s^t) = \pi(s^t/s^{t-1})\pi(s^{t-1})$$

or more generally, that

$$\pi(s^t) = \pi(s^t/s^{t-1})\pi(s^{t-1}/s^{t-2})...\pi(s_0)$$

The literature often writes the consumption of the household conditional on history s^t, $C_t(s^t)$, as simply C_t, thus suppressing the notation for the stochastic history of events. It is important to recognize (and remember!) that even if written in this way, as we will do on most other occasions in future chapters, C_t is a stochastic variable that depends on history, i.e. a particular sequence of events s^t for which the household has planned. It may, perhaps, seem clumsy to write $C_t(s^t)$ instead of simply $C(s^t)$. I do this to avoid confusion. The time subscript on C_t signifies that it refers to the consumption *decision* at time t which gives the consumer utility at time t, while s^t refers to the sequence of events (or history) we allow the consumption decision of the household at time

[1] Since it is assumed that s^t follows s^{t-1} then $\pi(s^{t-1}/s^t) = 1$

t to depend upon. In principle we could imagine that the household needs to make its consumption decision many periods in advance, e.g., two periods in advance, in which case we would write $C_t(s^{t-2})$. Such examples often arise when firms set prices some periods in advance.

The representative household maximizes:

$$E_0 \sum_{t=0}^{\infty} \beta^t u(C_t) = \max : \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(C_t(s^t))$$

The notation $E_t[X]$ denotes the condition expectation of a random variable X, conditioning upon the state of the world at date t.

3 Complete Markets

How we write the budget constraint depends on what kind of assets the household can purchase. In the last chapter we considered:

$$b_t(s^t) = V_t(s^t) + Y_t(s^t) - C_t(s^t) - T_t(s^t) \tag{1}$$

$$V_{t+1}(s^t) = [1 + r_t(s^t)]b_t(s^t) \tag{2}$$

where Y_t is stochastic endowment, b_t is a one-period, risk-free real bond (here a positive b_t is an *asset*), r_t is the one-period risk-free real rate, $C_t(s^t)$ is consumption, $T_t(s^t)$ is real taxes levied by the government. The variable $V_{t+1}(s^t)$ denotes the asset holding of the household at the beginning of period $t+1$. The assumption of last chapter was that only one type of asset was traded, namely one-period debt so that at time t, $V_{t+1}(s^t)$ is known.

The case of complete markets corresponds to the case in which there exist enough assets so that the household can achieve any pattern of returns across all states of the world. In the case of complete markets, then the asset holding of the household at the beginning of period t, $V_t(s^t)$, is a random variable. The assumption of complete markets means that the household can achieve any desired random pattern for $\{V_t(s^t)\}$ provided it is willing to pay the right price for it.

4 Arrow-Debreu Securities

For the household problem to have a solution there cannot exist any *arbitrage opportunities* – that is, asset prices have to be such that the household cannot make any transactions that give it risk-free profits. This means that the price

of any asset portfolio that has the random return $V_{t+1}(s^{t+1})$ must be the same, regardless what combination of assets generates it. Denote by $\frac{1}{1+r_t}$ the price of any portfolio that returns $V_{t+1}(s^{t+1}) = 1$ regardless of which state of the world s^{t+1} is reached. Then r_t is the real rate of return on a riskless one-period bond, if one is traded. Note that we can define a riskless real rate in the case of complete markets even if no riskless asset is actually traded in equilibrium. The same applies for all other assets.

For each possible state s^{t+1} at date $t+1$ let $p(s^{t+1})/(1+r_t)$ denote the price at time t of a portfolio that has the value $V_{t+1}(s^{t+1}) = 1$ if s^{t+1} occurs but zero otherwise. This is called an *Arrow-Debreu security*. Absence of arbitrage opportunities requires that:

$$\sum_{s^{t+1}} p(s^{t+1}) = 1$$

Similarly the absence of arbitrage opportunities requires that the price of any portfolio with random returns $V_{t+1}(s^{t+1})$ is equal to:

$$\sum_{s^{t+1}} \frac{p(s^{t+1})}{1+r_t} V_{t+1}(s^{t+1})$$

5 The Stochastic Discount Factor

Define a random variable $R_{t,t+1}$, the value of which in each possible state s^{t+1} is given by:

$$R_{t,t+1} \equiv \frac{p(s^{t+1})}{(1+r_t)\pi(s^{t+1}/s^t)}$$

where $\pi(s^{t+1}/s^t)$ is the conditional probability of reaching state s^{t+1}, conditional on being in state s^t.

The cost of an arbitrary portfolio $V_{t+1}(s^{t+1})$ at date t is then given by:

$$\sum_{s^{t+1}} \pi_t(s^{t+1}/s^t) R_{t,t+1}(s_{t+1}) V_{t+1}(s^{t+1}) = E_t R_{t,t+1} V_{t+1}(s^{t+1})$$

The random variable $R_{t,t+1}$ is called the (real) stochastic discount factor and can be used to price any arbitrary asset portfolio. The stochastic discount factor used to discount random income at dates more than one period in the future is $R_{t,T}$ and is given by:

$$R_{t,T} \equiv R_{t,t+1} * R_{t+1,t+2} * ... * R_{T-1,T}$$

It is customary to define $R_{t,t} \equiv 1$

Finally, observe that:

$$E_t R_{t,t+1} = \frac{1}{1+r_t}$$

Later on we will consider a *nominal* stochastic discount factor, $Q_{t,t+1}$, which can be used to price any stochastic stream of dollars. Denoting the nominal interest rate by i_t, the nominal stochastic discount factor satisfies:

$$E_t Q_{t,t+1} = \frac{1}{1+i_t}$$

6 Budget Constraint for Complete Markets

Under complete markets, the budget constraint is:

$$b_t(s^t) = V_t(s^t) + Y_t(s^t) - C_t(s^t) - T_t(s^t) \tag{3}$$

and

$$b_t(s^t) = E_t[R_{t,t+1} V_t(s^{t+1})] \tag{4}$$

Contrast (4) to (2). The second equation is saying that the price of the assets the household holds at the beginning of period $t+1$ at time t ($E_t[R_{t,t+1}V_t(s^{t+1})]$) has to be equal to what the household has left according to its budget constraint at the end of period t ($b_t(s^t)$). In contrast to (4), it is no longer the case that consumption pins down the asset holding of the household. Here there is also a portfolio decision by the household which we have not made explicit. What kind of asset will it hold? In general, the answer is indeterminate. There are several different classes of assets the household could in principle hold to support its consumption pattern.

7 Borrowing limit under Complete Markets

Under complete markets the borrowing limit is:

$$V_{t+1}(s^{t+1}) \geq -\sum_{T=t+1}^{\infty} E_{t+1} R_{t+1,T}\{Y_t(s^t) - T_t(s^t)\}$$

for any s^{t+1}. This condition says that for any realization of the state, the household can never take on more debt than the net present value of its future income.

8 Intertemporal Budget Constraint

We can combine the period budget constraint and the borrowing limit to yield the intertemporal budget constraint. Combine (3) and (4) to yield:

$$V_t(s^t) = E_t[R_{t,t+1}V_t(s^{t+1})] - Y_t(s^t) + C_t(s^t) + T_t(s^t) \tag{5}$$

Forward to yield:

$$V_{t+1}(s^{t+1}) = E_{t+1}[R_{t+1,t+2}V_{t+1}(s^{t+2})] - Y_{t+1}(s^{t+1}) + C_{t+1}(s^{t+1}) + T_{t+1}(s^{t+1})$$

Using the law of iterated expectations, and that $R_{t,t+2} = R_{t,t+1} * R_{t+1,t+2}$, we obtain:

$$\begin{aligned} V_t(s^t) = E_t[R_{t,t+2}V_{t+1}(s^{t+2})] \\ - E_tR_{t,t+1}Y_{t+1}(s^{t+1}) + E_tR_{t,t+1}C_{t+1}(s^{t+1}) \\ + E_tR_{t,t+1}T_{t+1}(s^{t+1}) - Y_t(s^t) + C_t(s^t) + T_t(s^t) \end{aligned} \tag{6}$$

Doing repeated substitution until period T and rearranging yields:

$$\sum_{s=0}^{T-1} R_{t,t+s}C_{t+s} + E_t[R_{t,T}V_T(s^T)] = E_t\sum_{s=0}^{T-1} R_{t,t+s}(Y_{t+s} - T_{t+s}) + V_t(s^t) \tag{7}$$

Apply the borrowing limit to period T, multiply both sides by $R_{t,T}$, take expectations at time t from both sides, and rearrange to yield:

$$E_tR_{t,T}V_T(s^T) + \sum_{s=T}^{\infty} E_tR_{t,s}\{Y_t(s^t) - T_t(s^t)\} \geq 0 \tag{8}$$

Add (8) to the right hand side of (7) to obtain the inequality:

$$\begin{aligned} &E_t\sum_{s=0}^{T-1} R_{t,t+s}C_{t+s} + E_t[R_{t,T}V_T(s^T)] \leq \\ &E_t\sum_{s=0}^{T-1} R_{t,t+s}(Y_{t+s}-T_{t+s}) + V_t(s^t) + E_tR_{t,T}V_T(s^T) - E_t\sum_{s=T}^{\infty} R_{t,s}\{Y_t(s^t)-T_t(s^t)\} \end{aligned} \tag{9}$$

Eliminating $E_t[R_{t,T}V_T(s^T)]$ from both sides, and combining the first and the last terms yields:

$$E_t\sum_{s=0}^{T-1} R_{t,t+s}C_{t+s} \leq E_t\sum_{s=0}^{\infty} R_{t,t+s}(Y_{t+s} - T_{t+s}) + V_t(s^t) \tag{10}$$

Since the left hand side holds for any $T-1$, it also holds in the limit, so that we obtain the intertemporal budget constraint:

$$E_t \sum_{s=0}^{\infty} R_{t,t+s} C_{t+s} \leq E_t \sum_{s=0}^{\infty} R_{t,t+s}(Y_{t+s} - T_{t+s}) + V_t(s^t) \tag{11}$$

We use the notation $IBC_t \geq 0$ to summarize the intertemporal budget constraint, i.e.

$$IBC_t = E_t \sum_{s=0}^{\infty} R_{t,t+s}(Y_{t+s} - T_{t+s}) + V_t(s^t) - E_t \sum_{s=0}^{\infty} R_{t,t+s} C_{t+s} \geq 0 \tag{12}$$

9 Deriving the Consumption Euler Equation under Complete Markets

The representative household maximizes:

$$\max \sum_{t=0}^{\infty} \beta^t \sum_{s^t} \pi\left(s^t\right) u(C_t(s_t))$$

subject the intertemporal budget constraint we just derived:

$$\sum_{t=0}^{\infty} \sum_{s^t} \pi(s^t) R_{0,t}(s^t) C_t(s^t) \leq \sum_{t=0}^{\infty} \sum_{s^t} \pi(s^t) R_{0,t}(s^t)(Y_t(s^t) - T_t(s^t)) + V_0$$

We can characterize the necessary conditions for optimality by solving the Lagrangian:

$$L_0 = \max \sum_{t=0}^{\infty} \beta^t \sum_{s_t} \pi\left(s^t\right) u(C_t(s^t)) + \lambda\{\sum \sum_{s_t} \pi(s^t) R_{0,t}(s^t)(Y_t(s^t) - T_t(s^t)) - a_0 - \sum \sum_{s_t} \pi(s^t) R_{0,t}(s^t) C_t(s^t)\}$$

The first-order conditions are given by:

$$\frac{\partial L_0}{\partial C_t(s^t)} = \beta^t \pi(s^t) u_c(C_t(s^t)) - \lambda \pi(s^t) R_{0,t}(s^t) = 0 \tag{13}$$

for all contingencies s^t or equivalently:

$$\beta^t \pi(s^t) u_c(C_t(s^t)) = \lambda \pi(s^t) R_{0,t}(s^t)$$

This condition holds for $C_t(s^t)$ and $C_t(s^{t+1})$. Dividing the two we obtain:

$$\frac{\beta^{t+1} \pi(s^{t+1}) u_c(C_{t+1}(s^{t+1}))}{\beta^t \pi(s^t) u_c(C_t(s^t))} = \frac{\pi(s^{t+1}) R_{0,t+1}(s^{t+1})}{\pi(s^t) R_{0,t}(s^t)} = \frac{\pi(s^{t+1}) R_{0,t}(s^t) R_{t,t+1}(s^{t+1})}{\pi(s^t) R_{0,t}(s^t)}$$

or:

$$R_{t,t+1}(s^{t+1}) = \frac{\beta u_c(C_{t+1}(s^{t+1}))}{u_c(C_t(s^t))} \tag{14}$$

Taking expectation of both sides, and noting that $E_t R_{t,t+1}(s^{t+1}) = \frac{1}{1+r_t}$ we obtain the standard consumption Euler Equation derived in last chapter:

$$\frac{1}{1+r_t} = E_t \frac{\beta u_c(C_{t+1}(s^{t+1}))}{u_c(C_t(s^t))} \tag{15}$$

Following the same steps, observe that more generally the stochastic discount factor for any time T and state s^T is:

$$R_{t,T}(s^T) = \frac{\beta^{T-t} u_c(C_T(s^T))}{u_c(C_t(s^t))} \tag{16}$$

which can be used to price multiperiod bonds. As the intertemporal budget constraint is an inequality, the optimal plan also satisfies a Kuhn-Tucker complementary slackness condition:

$$\lambda \geq 0,\ IBC_0 \geq 0 \text{ and } \lambda \times IBC_0 = 0 \tag{17}$$

Because the marginal utility of consumption is always positive, i.e. $u_c(C) > 0$ first-order condition (13) and condition (17) imply that the intertemporal budget constraint holds with equality, i.e.

$$IBC_0 = 0$$

is a necessary condition for household maximization. The economic interpretation of this result is that the household will always exhaust is budget constraint, because it will always prefer additional consumption.

10 Deriving the Transversality Condition

The transversality condition can be stated as:

$$\lim_{T->\infty} E_t R_{t,T} V_t = 0 \tag{18}$$

and this condition is part of the necessary condition for maximum for the representative household. Many textbooks, such as Stockey and Lucas (1987), omit proving the necessity of the transversality conditions for optimality and refer to it as a difficult problem. Kamihigashi (2002) contains one example of a relatively simple proof.

The exposition of the intertemporal budget constraint from last section, however, which follows Woodford (2003), allows for a simple way of showing the necessity of (18) as illustrated below

Proposition 1. *The intertemporal budget constraint (12) holds with equality* $\iff$ *Flow constraint (5) holds as well as (18)*

Proof. The proposition follows from forwarding the period budget constraint to yield:

$$E_t \sum_{s=t}^{T-1} R_{t,s} C_s + E_t R_{t,T} V_T = V_t + E_t \sum_{s=t}^{T-1} R_{t,s}(Y_s - T_s) \tag{19}$$

This equation is equation is equivalent to (12) holding with equality as $T \to \infty$ if $E_t R_{t,T} V_T = 0$. □

Discussion: Getting the intertemporal budget constraint (12) to hold with equality as a necessary condition for household maximization involved two steps. 1) The first step used the borrowing constraint, or no-Ponzi condition, to get to inequality (12). 2) Involved using the Kuhn-Tucker Theorem to obtain (17), and using the assumption that the marginal utility of consumption is always positive, which established that (12) had to hold with equality. Observe that step 1 involved preventing the household from accumulating debt ad infinum to pay off its own debt – i.e. preventing Ponzi schemes. The second step involved excluding the opposite possibility, i.e. that the household keeps accumulating assets, which would be sub-optimal as it could in that case increase its consumption stream, which it will always wish to do as long as its marginal utility is positive.

11 Pricing Assets

With the stochastic discount factor defined, we can price any assets. Consider an arbitrary asset that generates the income stream $\{d_t\}$. Consider the price of this asset at period t. It is:

$$p_t^d = E_t \sum_{T=t}^{\infty} R_{t,T} d_T$$

where the stochastic discount factor, $R_{t,T}$, is given by (14). An important observation is that the supply of the asset $\{d_t\}$ is irrelevant for the pricing of the asset. All that matters is the payoffs it generates $\{d_t\}$ as well as the stochastic discount factor, which does not depend on the supply of the asset either (see equation 14).

It is useful to observe that the price of an asset with income stream $\{d_t\}$ can similarly be written as:

$$p_t^d = d_t + E_t R_{t,t+1} p_{t+1}^d$$

i.e. the price of a security is equal to its dividend at time t plus the resale value of the security in the next period (pre-dividend payment).

12 The Ricardian Equivalence Result Generalized

This section shows how the Ricardian proposition from last chapter can be generalized. While the general notation – which is a special case of the notation introduced in Eggertsson and Woodford (2003) – may seem somewhat unwieldy, the idea is relatively straightforward. In the last chapter we showed that if the government issues only one-period, risk-free government bonds, then the time pattern of taxes is irrelevant. Here, instead, we allow for the possibility that the government purchases or sells any arbitrary quantity of assets. This includes debt of arbitrary duration with various risk characteristics. Essentially the proposition boils down to explicitly introducing notation for the government so that it can hold arbitrary amounts and types of financial securities and show that the pricing of these securities does not depend on how many or what type of securities the government holds. Essentially, this is because these securities are priced exactly in the same way as shown in equation (11), i.e. the price only depends on the net present value of the payout of the security. The strategy for setting up the proposition is the same as we saw in the last chapter, i.e. we will show that that the set of necessary and sufficient equilibrium conditions is independent of the policy functions that define the choices of the government.

At the beginning of period t we assume that the government holds a vector of various securities as given by the following form:

$$\omega_t = \omega(Y_t, \xi_t) \tag{20}$$

where the vector-valued function $\omega(.)$ has the property that its components sum up to 1 for all possible values of its arguments. In addition to output, we allow policy to depend on an exogenous shock ξ_t, which could for example represent a preference shock that enters directly in the utility function.

The payoffs of the securities the government holds are specified by an *exogenously given, state-contingent* vector $\mathbf{b}_t(s^t)$ and matrix $\mathbf{F}_t(s^t)$. A vector of asset holding $\mathbf{z}_{t-1}(s^t)$ at the end of period $t-1$ results in the delivery to the owner a quantity $\mathbf{b}_t'\mathbf{z}_{t-1}$ of the consumption good and a vector $\mathbf{F}_t\mathbf{z}_{t-1}$ of securities that may be traded in the period t asset market, each of which may depend on the state of the world, s^t, in period t. This specification allows us to consider a broad range of possibilities. A one-period riskless real bond, for example, has elements of zero in $\mathbf{F}_t(:,\mathbf{j})$ while $b_t(j)$ is the same in all states.[2] This notation

[2]It is a two-period riskless nominal pure discount bond if instead $b_t(j)$ is zero but $F_t(i,j)$=0 for all i's except when $i = k$. Instead $F_t(k,j) > 0$ is the same in all states of the world and the security the agent gets in period t (i.e. security k) is a one-period riskless real bond.

also allows for the possibility that the bonds issued by the government are risky, as long as that risk depends only on the exogenous states.[3]

The gross return $rr_t(j)$ on the jth asset between periods $t-1$ and t is then given by:

$$rr_t(j) = \frac{b_t(j) + \mathbf{q}_t' \mathbf{F}_t(:, \mathbf{j})}{q_{t-1}(j)} \tag{21}$$

where $\mathbf{q}_t$ is the vector of assets priced (after they have paid out dividends) in period-t trading, whose price is given by the net present value of the dividends they are expected to pay out:

$$\mathbf{q}_t' = E_t R_{t,t+1} \mathbf{b_{t+1}}' + \sum_{T \geq t+2} E_t R_{t,T} \mathbf{b}_T' \prod_{s=t+1}^{T-1} \mathbf{F}_s \tag{22}$$

The price of securities according to the equation above is, plainly, the implied stream of dividends of each security. The security is priced at time t after the period t dividend has been paid out. The first term represents the expected dividend payout of each security in period $t+1$. The second term captures future payouts. Recall that the matrix $\mathbf{F}_t$ captures what a given security implies in terms of other securities (which in turn have their own dividend payouts).

The government budget constraint is given by:

$$V_t = \mathbf{rr}_t' \omega_{t-1} V_{t-1} + G_t - T_t \tag{23}$$

where V_t denotes total government liabilities/asset holdings, i.e. it is the market value of the government debt/asset holding. The description of government behavior is completed by specifying a policy rule for taxes and spending:

$$T_t = T(Y_t, \xi_t) \tag{24}$$

$$G_t = G(Y_t, \xi_t) \tag{25}$$

To define an equilibrium we need to close the model by the aggregate resource constraint:

$$Y_t = C_t + G_t \tag{26}$$

If we substitute for $R_{t,T}$ in (22) and (18) using (16), we can define the equilibrium as follows:

Definition 2. *An equilibrium is a set of stochastic processes for* $\{r_t, \mathbf{rr}_t, \mathbf{q}_t, V_t, C_t\}$ *for* $t \geq 0$ *that satisfy (15), (18), (21), (22), (23) and (26) given the policy specification for* $\{\omega_{t,} T_{t,} G_{t,}\}$ *given by (20), (24) and (25), an exogenous process for* $\{Y_t\}$ *and an initial condition* ω_{-1} *and* V_{-1}.

[3] Importantly, however, we are excluding the possibility that the payouts depend on endogenous things, such as taxes or the level of debt or asset holding of the government

The list of equilibrium conditions above are necessary and sufficient for an equilibrium consistent with household maximization and rational expectations. Again we define a sustainable policy as a specification for $T(.)$ that ensures that the transversality condition (18) is satisfied for any realization of output and asset prices. We can now state a more general Ricardian Equivalence result.

Proposition 3. *Assume that fiscal policy is sustainable. In this case, the determination of* $\{r_t, \mathbf{rr}_t, \mathbf{q}_t, C_t\}$ *is independent of the specification of the functions* $T(Y_t, \xi_t)$ *and* $\omega(Y_t, \xi_t)$.

Proof. The proof of this is identical to the one from last chapter. We can simplify the list of necessary and sufficient conditions for equilibrium to:

$$\frac{1}{1+r_t} = E_t \frac{\beta u_c(Y_T - G(Y_T))}{u_c(Y_t - G(Y_t))}$$

$$\mathbf{q}_t' = E_t \sum_{T \geq t+1} \frac{\beta^{T-t} u_c(Y_T - G(Y_T))}{u_c(Y_t - G(Y_t))} \mathbf{b}_T' \prod_{s=t+1}^{T-1} \mathbf{F}_s \tag{27}$$

$$rr_t(j) = \frac{b_t(j) + \mathbf{q}_t' \mathbf{F}_t(:, \mathbf{j})}{q_{t-1}(j)}$$

$$V_t = \mathbf{rr}_\mathbf{t}' \omega_{\mathbf{t-1}} V_{t-1} + G_t - T_t \tag{28}$$

$$\lim_{T->\infty} E_t \beta^T u_c(Y_T - G(Y_T)) V_T = 0$$

For a given function $G(Y_t, \xi_t)$ and a stochastic process for Y_t, the first equation pins down r_t, as in the last chapter, independently of the function $T(Y_t, \xi_t)$ or $\omega(Y_t, \xi_t)$. All other assets are priced by the second equation. There, each is defined by elements of the exogenous *state-contingent* vector $\mathbf{a}_t$ and the matrix $\mathbf{F}_t$, and the stochastic discount factor $\frac{\beta^{T-t} u_c(Y_T - G(Y_T))}{u_c(Y_t - G(Y_t))}$, neither of which depends on $T(.)$ or $\omega(.)$. The first equation is a special case of the second one, included to make clear the connection with last chapter. The period returns of each of the assets are shown in the third equation, and these returns are similarly unaffected by the functions $T(.)$ and $\omega(.)$. The fourth equation, i.e. the budget constraint, shows the implied evolution of V_t for a given choice of the functions $\omega(.)$ and $T(.)$, but the value of V_t has no implication for the asset price equations above. By our definition of sustainable fiscal policy the last equation is always satisfied. □

This proposition generalizes the previous one to a considerable extent. Before, we only considered one-period risk-free debt. This proposition, in contrast, allows for the government to issue debt of any duration, moreover, it also allows for the possibility of risky debt, for example, the debt issued might have

the property that in some states of the world the government does not pay out anything.

An important feature of the assets considered, however, is that the payoffs of the securities being priced are given by an exogenous state vector $\mathbf{a}_t$ and matrix $\mathbf{F}_t$. Thus we are excluding *the possibility that government taxes, asset holdings or debt accumulation affects these payoffs*. If, for example, a tax cut increases the probability of government default and via this affects $\mathbf{a}_t$ and $\mathbf{F}_t$, the assumptions of the proposition are being violated.

A key insight of the proposition is that the relative supply of various assets has no effect on asset prices. Instead, all that matters is the state-contingent payoffs these assets imply. This will remain true even once we allow for endogenous production, price frictions and monetary policy. This will have fundamental implications once we start thinking about policies such as Quantiative Easing, a policy that corresponded to the Federal Reserve purchasing long-term government bonds in exchange for short-term bonds (Federal Reserve interest-bearing bank reserves). A similar proposition to this one will suggest that these policies were irrelevant, to the extent they did not change expectations about future interest rates. This is an issue we will study in some detail.

13 Discussion

TBA

14 Revisiting the one period bond case

Before suppressing the explicit notation taking account of different states of the world, as we do in coming chapters, it is worth revisiting briefly the derivation of the consumption Euler Equation from last chapter where we assumed that only one bond was traded.

We derived the consumption Euler equation by using the method of Lagrange. Writing the Lagrangian taking into account explicitly the state by state notation results in:

$$L_0 = \sum_{t=0}^{\infty} \beta^t \left\{ \sum_{s^t} \pi(s^t) \left[\begin{array}{c} u(C_t(s^t)) \\ +\phi_t(s^t)\{b_t(s^t) - (1 + r_{t-1}(s^{t-1}))b_{t-1}(s^{t-1}) - Y_t(s^t) + C_t(s^t) + T_t(s^t)\} \end{array} \right] \right\}$$

The household is choosing a consumption plan over different histories of events s^t, i.e. satisfying the first-order conditions:

$$\frac{\partial L_0}{\partial C_t(s^t)} = \beta^t \pi(s^t) u_c(C_t(s^t)) + \beta^t \pi(s^t) \phi_t(s^t) = 0 \tag{29}$$

$$\frac{\partial L_0}{\partial b_t(s^t)} = \beta^t \pi(s^t)\phi_t(s^t) - \beta^{t+1}(1 + r_t(s^t)) \sum_{s^{t+1}\backslash s^t} \pi(s^{t+1})\phi_{t+1}(s^{t+1}) = 0 \quad (30)$$

The notation $\sum_{s^{t+1}\backslash s^t}$ denotes the sum of all possible histories $\tilde{s}^{t+1}$ such that $\tilde{s}^t = s^t$.

Recall that we can write:

$$\pi(s^{t+1}) = \pi(s^{t+1}\backslash s^t)\pi(s^t) \quad (31)$$

Now we can rewrite the second first order condition as:

$$\phi_t(s^t) - \beta(1 + r_t(s^t)) \sum_{s^{t+1}\backslash s^t} \pi(s^{t+1}\backslash s^t)\phi_{t+1}(s^{t+1}) = 0$$

Observe that $\sum_{s^{t+1}\backslash s^t} \pi(s_{t+1}\backslash s_t)\phi_{t+1} = E_t\phi_{t+1}$ is the conditional expectation of ϕ_{t+1} at time t, given the state of the world in period t. It is very common to drop the notation involving states, which we will do from now on, and simply write:

$$\phi_t - \beta(1 + r_t)E_t\phi_{t+1} = 0$$

Combine this with the first-order condition (29) to yield the consumption Euler Equation:

$$u_c(C_t) = \beta(1 + r_t)E_t u_c(C_{t+1}) \quad (32)$$

The key thing to keep in mind, even if adopting the more suppressed notation above, is that each of the variables does depend on various states of the world.

15 Connection to the literature

The presentation in this chapter, and much of the notation, builds heavily on unpublished lecture notes by Michael Woodford that I studied in graduate school at Princeton in 1997. The closest published work that I have found that covered similar topics are Santos and Woodford (1997) and Woodford (1998) cited below, as well as Woodford's textbook.

References

[1] Manuel S. Santos and Michael Woodford. "Rational Asset Pricing Bubbles". *Econometrica* 65 (1997), pp. 19–58.

[2] Michael Woodford. "Doing Without Money: Controlling Inflation in a Post-Monetary World". *Review of Economic Dynamics* 1 (1998), pp. 173–219.